



WEST BENGAL STATE UNIVERSITY
B.Sc. Major 1st Semester Examination, 2025-26

CMSDSC101T/CMSDSCN101T-COMPUTER SCIENCE (MAJOR)

Time Allotted: 2 Hours

Full Marks: 50/40

(i) Students of Session 2023-24 & 2024-25: 50 Marks

(ii) Students Session 2025-2026 Onwards: 40 Marks

*The figures in the margin indicate full marks.
Candidates should answer in their own words and adhere to the word limit as practicable.
All symbols are of usual significance.*

- (a) Question No. 1 to 7 common for all students;
(b) Question No. 8 is only for the Students of Session 2023-24 & 2024-25

Answer Question No. 1 and any four questions from the rest

1. Answer any **four** questions from the following:

2×4 = 8

- What is file pointer?
- Write the difference between function and macros.
- Define inline function.
- Explain preprocessor directives in C.
- When do we need void key word?
- State the differences between library function and user defined function.
- What would be the output of following code?

```
main()
{ int a[5] = { 1,2,3};
printf("%d", a[4]);
```

Answer any four questions from the following

8×4 = 32

- Differentiate between null value and null pointer. What are the advantages and disadvantages of Array? Write a program to calculate the power of a user taken number without using system defined function. 2+2+4
- What are the contents of the command line arguments argc and argv? What is the importance of 'break' in switch case? Differentiate between malloc() and calloc(). What do you mean by function prototype? 2+2+2+2
- Specify the difference between switch case statement & loop. Why do we need loop structure in programming? Explain the drawback of switch case. Write a C program to print the following pattern: 2+1+2+3

```
1
12
123
1234
```

5. (a) Explain call by value and call by reference with examples.
(b) Write a C program to store and display 5 integers in an array.
(c) What is recursion? Write a recursive function to find factorial.
6. (a) Explain pointer arithmetic with example.
(b) Write a C program to open and write to a file.
(c) Differentiate between get() and gets() functions.
7. (a) What are keywords and identifiers? Give examples.
(b) Explain arithmetic and relational operators with examples.
(c) Evaluate: $(10 + 5) * 2 - 8 / 4$.

2
3
3
2
3
3
2
3
3

ADDITIONAL FOR SUPPLEMENTARY CANDIDATES ONLY
For Students of Session 2023-24 & 2024-25 only (10 marks)

8. (a) What is the use of %d in C?
(b) What is the difference between abs() and fabs() functions?
(c) What is the purpose of printf() and scanf() functions in C program?
(d) Write a simple example of a structure in C Language.

2
2
3
3

—x—